

Leverage and Influence

2026-04-17

Introduction

Leverage and influence diagnostics identify observations whose removal would substantially change a regression fit. The two concepts are distinct: leverage measures how unusual an observation is in the predictor space, influence measures how much it actually affects the fitted coefficients. High leverage without a large residual is harmless; high residual without leverage produces a poor fit but not influential coefficients; only the combination of both produces an observation that disproportionately drives the regression.

Prerequisites

A working understanding of residual diagnostics, the hat matrix, and the leave-one-out perspective on regression.

Theory

Hat values h_{ii} are the diagonal of the projection matrix $H = X(X^\top X)^{-1}X^\top$. They measure each observation's leverage in the predictor space; the trace $\sum h_{ii} = p + 1$ (number of parameters), so the average is $(p + 1)/n$. Conventional threshold: $h_{ii} > 2(p + 1)/n$ is high leverage.

DFBETA for coefficient j and observation i : standardised change in $\hat{\beta}_j$ when observation i is removed. Threshold: $|\text{DFBETA}_{ij}| > 2/\sqrt{n}$.

Cook's distance: combines residual and leverage into a single influence summary; thresholds $D_i > 4/n$ or $D_i > 1$.

DFFITS: standardised change in fitted value at observation i ; analogous to DFBETA but for predictions rather than coefficients.

Assumptions

Classical OLS assumptions; the diagnostics above are linear-regression-specific. For GLMs, `influence.measures()` provides analogues based on one-step approximations.

R Implementation

```
library(car)

fit <- lm(mpg ~ wt + hp, data = mtcars)

# Hat values
h <- hatvalues(fit)
which(h > 2 * (length(coef(fit))) / nrow(mtcars))

# DFBETAs
dfb <- dfbetas(fit)
which(apply(abs(dfb), 1, max) > 2 / sqrt(nrow(mtcars)))
```

```
# Cook's distance
cooks <- cooks.distance(fit)
which(cooks > 4 / nrow(mtcars))

# Influence plot
influencePlot(fit)
```

Output & Results

Each diagnostic flags candidate observations. `car::influencePlot()` produces a single integrated plot with leverage on the x-axis, studentised residual on the y-axis, and point size proportional to Cook’s distance — the canonical three-way diagnostic.

Interpretation

A reporting sentence: “Influence diagnostics flagged three observations as candidates for further inspection (Maserati Bora: high leverage and large residual; Cadillac Fleetwood, Lincoln Continental: high leverage with moderate residuals); refitting without these observations changed the coefficient on weight by less than 0.3 SE units, with substantive conclusions unchanged.” Sensitivity analyses are the right way to report influence diagnostics; pointing out flagged observations without consequence checks adds noise rather than signal.

Practical Tips

- Leverage is determined by the predictor matrix alone; outliers in the response are residual problems, not leverage.
- Only the combination of leverage and large residual produces influence; investigate flagged points before removing them.
- Never delete influential observations without investigating their cause — they may be the most informative data points or signal genuine measurement errors.
- Report sensitivity analyses with influential points removed; transparency about robustness builds reader confidence.
- For GLMs, `influence.measures(fit)` returns DFBETAs, hat values, and Cook’s-distance equivalents based on one-step approximations.
- Robust regression (`MASS::rlm`, `robustbase::lmrob`) automatically downweights influential observations; useful when influential points are genuine and should be retained but should not dominate the fit.

R Packages Used

Base R `hatvalues()`, `dfbetas()`, `dffits()`, `cooks.distance()`, `influence.measures()`; `car::influencePlot()` for the three-way diagnostic visualisation; `performance::check_outliers()` for tidy diagnostics; `MASS::rlm()` and `robustbase::lmrob()` for robust regression alternatives.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression

- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI